

Web and Multimedia Technologies

Lab class 8 – PHP + MySQL/MariaDB

Instructions to use the laboratory computers (room B2/B4):

1. If the computer is off, start it with Windows (all computers have a Linux-Windows dual boot)
2. Your username is *cl* followed by the six digits of your student ID; your password is *inf* followed by the last two digits of your student ID (e.g., if the student ID is 123456, the username is *cl123456* and the password is *inf56*)
3. At the end of the class, please log out

Steps (Apache web server, PHP application server, and MySQL or MariaDB DBMS are needed):

1. Start the Apache, PHP, and MySQL (or MariaDB) services (e.g., by starting the *Wampserver* or equivalent) and verify that the *Apache* web server is working correctly by loading the page *localhost* (or *127.0.0.1*): the default home page should be displayed
2. The “document root” of the web server can be changed by modifying the corresponding entry in the `httpd.conf` configuration file (e.g., with the Wampserver: click the small ‘W’ icon in the Windows status bar and, in the menu that is displayed, choose *Apache* → *httpd.conf*; the default root is `c:\install directory>\www`). In the root directory, create the folder “wmt8”
3. Open the *phpMyAdmin* configuration page (it may be reached through a link in the *localhost* page or through a menu of the Wampserver or other installations). The default username is always *root*, while the default password is the empty string for Wampserver and XAMPP, and *root* for MAMP); select *English* as a language
4. Click the *Databases* tab
5. Create a new database named “mydb”
6. Click *mydb* on the left and create a new table “users” with 5 fields (columns): *id* (INT, length 11, auto increment, primary key), *name* (VARCHAR, length 100), *city* (VARCHAR, length 100), *email* (VARCHAR, length 30), *year* (SMALLINT, length 6). Click *Save* to save the table
7. Enter a couple of rows into the table with the following SQL code (*SQL* tab):

```
INSERT INTO users (id,name,city,email,year) VALUES (1, 'Leonardo da Vinci',  
'Vinci', 'leo@dvc.com', 1452);  
INSERT INTO users (id,name,city,email,year) VALUES (2, 'Giuseppe Verdi',  
'Busseto', 'giuse@greens.com', 1813);
```


(then press *Go*)
8. Create a new table named “usrpwd”, with two fields: *username* (VARCHAR, length 15, primary key) and *password* (VARCHAR, length 10)
9. Download the zip file <https://vision.unipv.it/wmt/lab/php/phpfiles.zip> (to the folder “wmt8” in the root directory), which contains the files `passwords.php`, `login.php`, `reg.php`, and `db.php`
10. Analyze the first three files (`passwords.php`, `login.php`, and `reg.php`)
11. In “wmt8”, create the file `registration.html` with the following content (all files will be placed in “wmt8”):

```
<!DOCTYPE html>  
<html>  
  <head>  
    <meta charset="utf-8">  
    <title>Registration</title>  
  </head>  
  <body>  
    <h1>Registration</h1>  
    <form name="regform" action="reg.php" method="post">  
      Username <input type="text" name="usrn" id="usrn"><br><br>  
      Password <input type="password" name="pass" id="pass"><br><br>  
      <input type="submit" value="Submit">  
    </form>  
  </body>  
</html>
```

12. Create the file `dbinteraction.php` with the following content:

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>PHP + MySQL trial</title>
  </head>
  <body>
    <?php
      session_start();
      include("passwords.php");
      check_logged();
    ?>
    <h1>Interaction with a remote database</h1>
    <form name="f1" action="db.php" method="post">
      1- Show database content
      <input type="hidden" name="action" id="action" value="1">
      <input type="submit" value="Ok">
    </form>
    <form name="f2" action="db.php" method="post">
      2- Show data of the user whose name is
      <input type="hidden" name="action" id="action" value="2">
      <input type="text" name="n" id="n">
      <input type="submit" value="Ok">
    </form>
    <form name="f3" action="db.php" method="post">
      3- Enter a new record
      <input type="hidden" name="action" id="action" value="3">
      <input type="text" name="n" id="n" value="Name" size="10">
      <input type="text" name="c" id="c" value="City" size="10">
      <input type="text" name="e" id="e" value="E-mail" size="10">
      <input type="text" name="b" id="b" value="Birth year" size="10">
      <input type="submit" value="Ok">
    </form>
    <form name="f4" action="db.php" method="post">
      4- Delete the record of the user whose name is
      <input type="hidden" name="action" id="action" value="4">
      <input type="text" name="n" id="n">
      <input type="submit" value="Ok">
    </form>
  </body>
</html>
```

13. Analyze the file `db.php`, which implements the interaction with the database
14. The home page is `localhost/wmt8/dbinteraction.php`: after the login (and registration), it will be possible to perform simple operations on the database through the following interface:

Interaction with a remote database

1- Show database content

2- Show data of the user whose name is

3- Enter a new record

4- Delete the record of the user whose name is